Laplace's Equation

following Evan's *Partial Differential Equations* §2.2

The following defines one of the most important partial differential equations.

DEF 1. The Laplacian, denoted Δ, on $\mathbb{R}^n$ with standard coordinates $(x_1, ..., x_n)$ is defined,

$$\Delta = \sum_{i=1}^{n} \frac{\partial^2}{\partial x_i^2}$$

It is also sometimes denoted ∇^2 or $\nabla \cdot \nabla$. However we reserve the former notation for the Hessian.

DEF 2. Laplace's equation, and its inhomogeneous counterpart Poisson's equation, is respectively,

$$\Delta u = 0, \qquad -\Delta u = f$$

Where the function is $u \colon \bar{U} \to \mathbb{R}$ for a given open set $U \subseteq \mathbb{R}^n$, and in Poisson's equation a given inhomogeneous function $f \colon U \to \mathbb{R}$. The C^2 function u satisfying Laplace's equation is said to be 'harmonic'.

The following is a key characteristics of harmonic functions.

THM 1. HARMONIC FUNCTIONS ARE RADIALLY SYMMETRIC

Let $\mathbf{O} \in \mathbb{R}^{n \times n}$ be an orthogonal matrix. If u is harmonic, then so too is $v \colon \mathbf{x} \mapsto u(\mathbf{O}\mathbf{x})$.

Proof.

We compute the Laplacian of v using the chain-rule on the coordinates,

$$\Delta v = \sum_{i=1}^{n} \frac{\partial^2 v}{\partial x_i^2} = \sum_{i=1}^{n} \frac{\partial^2}{\partial x_i^2}[u \circ \mathbf{O}] = \sum_{i=1}^{n} \frac{\partial}{\partial x_i}\left[\frac{\partial u}{\partial x_i} \circ \mathbf{O} \cdot \mathbf{O}\right] = \mathbf{O} \sum_{i=1}^{n} \frac{\partial^2 u}{\partial x_i^2} \circ \mathbf{O} \cdot \mathbf{O} = \mathbf{O}^2 \sum_{i=1}^{n} \frac{\partial^2 u}{\partial x_i^2} = 0$$

■

THEOREM 1. motivates that finding any solution of Laplace's equation should be radially symmetric, i.e. functions purely of $r = |\mathbf{x}|$. We first start on the domain $\mathbb{R}^n$ whereby $|\mathbf{x}| = \sqrt{x_1^2 + \cdots + x_n^2}$. In particular by differentiating,

$$\frac{\partial |\mathbf{x}|}{\partial x_i} = \frac{1}{2}(x_1^2 + \cdots + x_n^2)^{-1/2} \cdot 2x_i = \frac{x_i}{r}$$

Therefore if we parameterize functions radially as in $u(\mathbf{x}) =: v(r)$ then by the chain-rule,

$$\frac{\partial u}{\partial x_i} = v'(r)\frac{x_i}{r}, \qquad \frac{\partial^2 u}{\partial x_i^2} = v''(r)\frac{x_i^2}{r^2} + v'(r)\left(\frac{1}{r} - \frac{x_i^2}{r^3}\right)$$

Consequently the Laplacian written radially is,

$$\Delta u = v''(r) + \frac{n-1}{r} v'(r)$$

And so a radial function is harmonic *iff*,

$$\frac{1-n}{r} = \frac{v''}{v'} = (\ln v')'$$

This is a separable ODE with solution,

$$v(r) = \begin{cases} a \ln r + b, & n = 2 \\ a/r^{n-2} + b, & n \geq 3 \end{cases}, \qquad v'(r) = \frac{a}{r^{n-1}}$$

For constants $a, b \in \mathbb{R}$. This is the fundamental solution to Laplace's equation up to a constant factor.

Namely the constant factor is as follows.

Def 3. The fundamental solution to Laplace's equation is,

$$\Phi(\mathbf{x}) := \begin{cases} -\frac{1}{2\pi} \ln |\mathbf{x}|, & n = 2 \\ \frac{1}{n(n-2)\omega_n} \frac{1}{|\mathbf{x}|^{n-2}}, & n \geq 3 \end{cases}$$

Observe,

$$|\nabla |\mathbf{x}|| = \left| \left(\frac{x_1}{|\mathbf{x}|}, ..., \frac{x_n}{|\mathbf{x}|} \right) \right| = \sqrt{\frac{1}{|\mathbf{x}|^2} \sum_{i=1}^{n} x_i^2} = 1$$

Which gives rise (through the chain rule) to the simple estimates,

$$|\nabla \Phi| \leq \frac{C}{|\mathbf{x}|^{n-1}}, \qquad |\nabla^2 \Phi| \leq \frac{D}{|\mathbf{x}|^n}$$

For some constants $C, D > 0$.

Using the fundamental solution we may solve Poisson's equation for a given f. Evidently $\Phi(\mathbf{x})$ is harmonic, and translations are rigid transformations so $\Phi(\mathbf{x} - \mathbf{y})$ for an arbitrary $\mathbf{y} \in \mathbb{R}^n$ is harmonic as a function of $\mathbf{x}$. Lastly multiplication by a factor $\Phi(\mathbf{x} - \mathbf{y}) f(\mathbf{y})$ preserves harmonicity as a function of $\mathbf{x}$. We would like this to be so for all $\mathbf{y}$ and in all direction, so we integrate over $\mathbb{R}^n$. This leads to the following result.

Thm 2. General Solution to Poisson's Equation on $\mathbb{R}^n$

Let f be continuously differentiable with compact support on $\mathbb{R}^n$. The solution to Poisson's equation is,

$$u(\mathbf{x}) = \int_{\mathbb{R}^n} \Phi(\mathbf{x} - \mathbf{y}) f(\mathbf{y}) \mathrm{d}\mathbf{y}$$

Proof.

Given $\Phi(\mathbf{x} - \mathbf{y})$ has a singularity when $\mathbf{x} = \mathbf{y}$ we must separate this into two cases. Namely first that u is first well-defined on all $\mathbb{R}^n$, and then that it solves Poisson's equation.

By a change-of-variables $\mathbf{y} = \mathbf{x} - \mathbf{y}$ we get,

$$\int_{\mathbb{R}^n} \Phi(\mathbf{x}-\mathbf{y}) f(\mathbf{y}) \mathrm{d}\mathbf{y} = \int_{\mathbb{R}^n} \Phi(\mathbf{y}) f(\mathbf{x}-\mathbf{y}) \mathrm{d}\mathbf{y}$$

Thereby computing the partial derivative of u is to consider,

$$\frac{\partial u}{\partial x_i} = \frac{u(\mathbf{x}+h\mathbf{e}_i) - u(\mathbf{x})}{h} = \int_{\mathbb{R}^n} \Phi(\mathbf{y}) \frac{f(\mathbf{x}-\mathbf{y}+h\mathbf{e}_i) - f(\mathbf{x}-\mathbf{y})}{h} \mathrm{d}\mathbf{y} \quad \rightrightarrows \int_{\mathbb{R}^n} \Phi(\mathbf{y}) \frac{\partial f}{\partial x_i}(\mathbf{x}-\mathbf{y}) \mathrm{d}\mathbf{y}$$

Where the uniform convergence follows from convergence on a compact domain. Similarly,

$$\partial_{i,j} u = \int_{\mathbb{R}^n} \Phi(\mathbf{y}) \partial_{i,j} f(\mathbf{x}-\mathbf{y}) \mathrm{d}\mathbf{y}$$

All terms of the integral are continuous w.r.t. $\mathbf{x}$, so the integral is continuous as a function of $\mathbf{x}$. Thus u is C^2 on $\mathbb{R}^n$.

For the singularity at the origin let $\varepsilon > 0$ and consider the integrals,

$$\Delta u(\mathbf{x}) = \int_{|\mathbf{y}|<\varepsilon} \Phi(\mathbf{y}) \Delta_x f(\mathbf{x}-\mathbf{y}) \mathrm{d}\mathbf{y} + \int_{|\mathbf{y}|>\varepsilon} \Phi(\mathbf{y}) \Delta_x f(\mathbf{x}-\mathbf{y}) \mathrm{d}\mathbf{y}$$

The first integral may be estimated a la Cauchy estimates,

$$\left| \int_{|\mathbf{y}|<\varepsilon} \Phi(\mathbf{y}) \Delta_x f(\mathbf{x}-\mathbf{y}) \mathrm{d}\mathbf{y} \right| \leq |\mathbb{B}^n_\epsilon| ||\nabla^2 f||_\infty \sup_{|\mathbf{y}|<\epsilon} \Phi(\mathbf{y}) \leq \begin{cases} C\varepsilon^2 \ln \varepsilon, & n = 2 \\ C\varepsilon^2, & n \geq 3 \end{cases}$$

The second integral may be seen using Green's second identity,

$$\int_{|\mathbf{y}|>\varepsilon} \Phi(\mathbf{y}) \Delta_{\mathbf{x}} f(\mathbf{x}-\mathbf{y}) \mathrm{d}\mathbf{y} = \oint_{|\mathbf{y}|=\varepsilon} \Phi(\mathbf{y}) \frac{\partial f}{\partial \mathbf{n}}(\mathbf{x}-\mathbf{y}) - \frac{\partial \Phi}{\partial \mathbf{n}}(\mathbf{y}) f(\mathbf{x}-\mathbf{y}) \, \mathrm{d}S_{\mathbf{y}} + \underbrace{\int_{|\mathbf{y}|>\varepsilon} f(\mathbf{x}-\mathbf{y}) \Delta_{\mathbf{y}} \Phi(\mathbf{y}) \, \mathrm{d}\mathbf{y}}_{0}$$

Compact support of f again allow us to estimate the first surface integral,

$$\left| \oint_{|\mathbf{y}|=\varepsilon} \Phi(\mathbf{y}) \frac{\partial f}{\partial \mathbf{n}}(\mathbf{x}-\mathbf{y}) \mathrm{d}\mathbf{y} \right| \leq |\mathbb{B}^n_\varepsilon| ||\nabla f||_\infty \sup_{|\mathbf{y}|=\varepsilon} \Phi(\mathbf{y}) = \begin{cases} C\varepsilon \ln \varepsilon, & n = 2 \\ C\varepsilon, & n \geq 3 \end{cases}$$

For the second surface integral observe generally that $\nabla \Phi(\mathbf{y}) = -\mathbf{y}/(n|\mathbb{B}^n_\varepsilon||\mathbf{y}|^n)$ and specifically that $\mathbf{n} = -\mathbf{y}/\varepsilon = -\mathbf{y}/|\mathbf{y}|$ on $\{|\mathbf{y}| = \varepsilon\}$. Therefore $\partial \Phi / \partial \mathbf{n}(\mathbf{y}) = 1/n\omega_n \varepsilon^{n-1} = 1/|\partial \mathbb{B}^n_\varepsilon|$. Thus,

$$\oint_{|\mathbf{y}|=\varepsilon} \frac{\partial \Phi}{\partial \mathbf{n}}(\mathbf{y}) f(\mathbf{x}-\mathbf{y}) \, \mathrm{d}S_{\mathbf{y}} = \frac{1}{|\partial \mathbb{B}^n_\varepsilon|} \oint_{|\mathbf{y}|=\varepsilon} f(\mathbf{x}-\mathbf{y}) \, \mathrm{d}S_{\mathbf{y}} \to f(\mathbf{x})$$

Using continuity of f. Thus in the limit $\varepsilon \to 0$ we indeed obtain,

$$\Delta u(\mathbf{x}) \to -f(\mathbf{x})$$

$\blacksquare$

This may be put more succintly via,

DEF 4. Let $\delta_{\mathbf{0}}$ denote the Dirac measure on $\mathbb{R}^n$ of unit mass at $\mathbf{0}$. That is, it satisfies,

$$\int_{\mathbb{R}^n} f(\mathbf{x})\delta_{\mathbf{0}}\,\mathrm{d}\mathbf{x} = f(\mathbf{0})$$

Then our previous computation becomes, with the singularity at $\Phi(\mathbf{0})$,

$$\Delta u(\mathbf{x}) = \int_{\mathbb{R}^n} \Delta_{\mathbf{x}}\Phi(\mathbf{x}-\mathbf{y})f(\mathbf{y})\,\mathrm{d}\mathbf{y} = -\int_{\mathbb{R}^n} \delta_{\mathbf{x}} f(\mathbf{y})\,\mathrm{d}\mathbf{y} = f(\mathbf{x})$$

We now consider the mean-value properties of harmonic functions. Let $U \subseteq \mathbb{R}^n$ be an arbitrary open set.

THM 3. MEAN-VALUE FORMULAS FOR LAPLACE'S EQUATION

Let $u \in \mathrm{C}^2(U)$ be harmonic. Then,

$$u(\mathbf{x}) = \frac{1}{|\partial B_r^{\mathbf{x}}|}\oint_{\partial B_r^{\mathbf{x}}} u\,\mathrm{d}S = \frac{1}{|B_r^{\mathbf{x}}|}\int_{B_r^{\mathbf{x}}} u\,\mathrm{d}V$$

For any ball $B_r^{\mathbf{x}} \subseteq U$.

Proof.

To prove the first equality define and by a translation and a contraction change-of-variables we have,

$$\phi(r) = \frac{1}{|\partial B_r^{\mathbf{x}}|}\oint_{\partial B_r^{\mathbf{x}}} u\,\mathrm{d}S = \frac{1}{|\partial\mathbb{B}^n|}\oint_{\partial\mathbb{B}^n} u(\mathbf{x}+r\mathbf{z})\,\mathrm{d}S_{\mathbf{z}}$$

Differentating the latter,

$$\phi'(r) = \frac{1}{|\partial\mathbb{B}^n|}\oint_{\partial\mathbb{B}^n} \nabla u(\mathbf{x}+r\mathbf{z})\cdot\mathbf{z}\,\mathrm{d}S_{\mathbf{z}} = \frac{1}{|\partial\mathbb{B}^n|}\oint_{\partial\mathbb{B}^n} \nabla u\cdot\mathrm{d}\mathbf{S}$$

Therefore using the divergence theorem,

$$\oint_{\partial\mathbb{B}^n} \nabla u\cdot\mathrm{d}\mathbf{S} = \int_{\mathbb{B}^n} \Delta u = 0$$

So that ϕ must be a constant function. In particular by computing its value at $r = 0$ (using continuity of ϕ and u) we find that $\phi = u(\mathbf{x})$.

For the second equality we utilize a polar transformation for,

$$\int_{B_r^{\mathbf{x}}} u\,\mathrm{d}V = \int_0^r \int_{\partial B_{\mathbf{x},s}} u\,\mathrm{d}S\,\mathrm{d}s = u(\mathbf{x})n\omega_n\int_0^r s^{n-1}\,\mathrm{d}s = \omega_n r^n u(\mathbf{x})$$

■

We also have the converse to this.

THM 4. MEAN-VALUE PROPERTY IS HARMONIC

If $u \in \mathrm{C}^2(U)$ satisfies,

$$u(\mathbf{x}) = \frac{1}{|\partial B_r^{\mathbf{x}}|}\oint_{\partial B_r^{\mathbf{x}}} u\,\mathrm{d}S$$

Then u is harmonic.

Proof.

Assume for contradiction the Laplacian of u is nonzero somewhere. Then by continuity there exists a ball $B_r^{\mathbf{x}} \subseteq U$ about this point s.t. $\Delta u > 0$. By the previous computation we would then have,

$$0 = \phi'(r) = \frac{1}{|\partial B_r^{\mathbf{x}}|} \int_{B_r^{\mathbf{x}}} \Delta u \, \mathrm{d}V > 0$$

A contradiction.

■

The mean-value theorem may be used to prove uniqueness of solutions to Laplace's equation. For this we have a extrema result. Let $U \subseteq \mathbb{R}^n$ be an arbitrary open set.

Thm 5. Strong Maximum Principle

Let $u \in \mathrm{C}^2(U) \cap \mathrm{C}(\bar{U})$. Then,

$$\max_{\bar{U}} u = \max_{\partial U} u$$

And if U is connected and there is a point $\mathbf{x}_0 \in U$ s.t. $u(\mathbf{x}_0) = \max_{\bar{U}} u$, then u is constant on U.

Proof.

Let $\mathbf{x}_0 \in U$ attain the maximum of u on $\bar{U}$, i.e. $\max_{\bar{U}} u = u(\mathbf{x}_0)$. In a ball of radius r about $\mathbf{x}_0$ we have by the mean-value property, and by a Cauchy estimate,

$$\max_{\bar{U}} u = u(\mathbf{x}_0) = \frac{1}{B_r^{\mathbf{x}_0}} \int_{B_r^{\mathbf{x}_0}} u \, \mathrm{d}V \leq \max_{\bar{U}} u$$

With equality *iff* u is constant on $B_r^{\mathbf{x}_0}$, which we must therefore satisfy. In the case U is connected the set $\{u(\mathbf{x}) = \max_{\bar{U}} u\}$ is an open and closed subset, so is all U. Therefore u is constant on U in this case. Otherwise the values of u in the interior are strictly less than its values on the boundary.

■

Note by $u \to -u$ that this also says that the minimum of u may only be attained on the boundary unless it is constant. Therefore the statement of Theorem 5. may be strengthened to say that the extrema of u exist only on the boundary of its domain unless it is constant. Theorem 5. in particular allows us to conclude that for the problem,

$$\begin{cases} \Delta u(\mathbf{x}) = 0, & \mathbf{x} \in U \\ \quad u(\mathbf{x}) = g(\mathbf{x}), & \mathbf{x} \in \partial U \end{cases}$$

Where U is connected and $u \in \mathrm{C}^2(U) \cap \mathrm{C}(\bar{U})$, if g is generally nonnegative and positive somewhere on ∂U, then u is positive everywhere. This is by the strong extrema principle: g is lower-bounded by zero, and this may only be on the boundary. Since g is nonconstant, u is nonconstant, so is strictly greater than zero. This observation gives us the following general result.

Thm 6. Uniqueness of Solutions to Poisson's Equation

Let $g \in \mathrm{C}(\partial U)$ and $f \in \mathrm{C}(U)$. Then there is at most one solution $u \in \mathrm{C}^2(U) \cap \mathrm{C}(\bar{U})$ to the boundary-value problem,

$$\begin{cases} -\Delta u(\mathbf{x}) = f(\mathbf{x}), & \mathbf{x} \in U \\ \quad u(\mathbf{x}) = g(\mathbf{x}), & \mathbf{x} \in \partial U \end{cases}$$

Proof.

Assume for contradiction u, u' were both solutions to the problem. Consider $w := u - u'$ and observe that in the interior $\Delta w = \Delta u - \Delta u' = f - f = 0$ and on the boundary $w = u - u' = g - g = 0$. Therefore w solves the problem,

$$\begin{cases} \Delta w(\mathbf{x}) = 0, & \mathbf{x} \in U \\ w(\mathbf{x}) = 0, & \mathbf{x} \in \partial U \end{cases}$$

By the strong extrema principle the values of w are $0 \leq w(\bar{U}) \leq 0$, so $w = 0$ is constant on $\bar{U}$. Therefore $u - u' = 0 \Rightarrow u = u'$.

■

The next major result for harmonic functions is that they are actually smooth, i.e. infinitely-differentiable. This is a regularity theorem for harmonic functions.

Thm 7. Harmonic Functions are Smooth

If $u \in \mathrm{C}(U)$ satisfies the mean-value property for each ball in U, then $u \in \mathrm{C}^\infty(U)$.

Proof.

Let η be the standard mollifier, i.e.,

$$\eta(\mathbf{x}) = \begin{cases} C \exp\left(\frac{1}{|\mathbf{x}|^2 - 1}\right), & |\mathbf{x}| < 1 \\ 0, & |\mathbf{x}| \geq 1 \end{cases}$$

Note η is radial, and we have its contraction to the ball of radius ε denoted η_ε. Denote $u^\varepsilon := \eta_\varepsilon * u$ on $U_\varepsilon = \{\mathbf{x} \in U \mid d(x, \partial U) > \varepsilon\}$. By convolution with the mollifier $u^\varepsilon \in \mathrm{C}^\infty(U)$. We claim that $u = u^\varepsilon$ on U_ε. Let $\mathbf{x} \in U_\varepsilon$ be arbitrary and,

$$\begin{aligned}
u^\varepsilon(\mathbf{x}) &= \int_U \eta_\varepsilon(\mathbf{x} - \mathbf{y}) u(\mathbf{y}) \, \mathrm{d}V_{\mathbf{y}} \\
&= \frac{1}{\varepsilon^n} \int_{B_r^{\mathbf{x}}} \eta\left(\frac{|\mathbf{x} - \mathbf{y}|}{\varepsilon}\right) u(\mathbf{y}) \, \mathrm{d}V_{\mathbf{y}} \\
&= \frac{1}{\varepsilon^n} \int_0^\varepsilon \eta\left(\frac{r}{\varepsilon}\right) \left(\int_{\partial B_r^{\mathbf{x}}} u \, \mathrm{d}S\right) \mathrm{d}r && \text{(Polar Transform about } \mathbf{x}\text{)} \\
&= \frac{u(\mathbf{x})}{\varepsilon^n} \int_0^\epsilon \eta\left(\frac{r}{\varepsilon}\right) n\omega_n r^{n-1} \, \mathrm{d}r && \text{(Mean-Value Property)} \\
&= u(\mathbf{x}) \int_{\mathbb{B}_\varepsilon^n} \eta_\varepsilon \, \mathrm{d}V && \text{(Inverse Polar Transform)} \\
&= u(\mathbf{x})
\end{aligned}$$

Given ε is arbitrary this shows u is smooth on U.

■

Using the mean-value property we also have several estimates on harmonic functions.

Thm 8. Cauchy Estimate on Derivatives

Let u be harmonic on U. Then,

$$\left|\frac{\partial^{\boldsymbol{\alpha}} u}{\partial x_{\boldsymbol{\alpha}}}\right| \leq \frac{C_k}{r^{n+k}} \|u\|_{\mathrm{L}^1(B_r^{\mathbf{x}_0})}$$

For each ball $B_r^{\mathbf{x}_0} \subseteq U$. The constant is,

$$C_k = \frac{(2^{n+1}nk)^k}{\omega_n}$$

Proof.

We prove this by induction on k. The $k=0$ case is given by the mean-value property,

$$|u(\mathbf{x}_0)| = \left|\frac{1}{|B_r^{\mathbf{x}_0}|}\int_{B_r^{\mathbf{x}_0}} u \,\mathrm{d}V\right| \leq \frac{1}{\omega_n r^n}\int_{B_r^{\mathbf{x}_0}} |u| \,\mathrm{d}V = \frac{C_0}{r^n}\|u\|_{\mathrm{L}^1(B_r^{\mathbf{x}_0})}$$

For $k=1$ by smoothness of harmonic functions we may commute $0 = \partial_i \Delta u = \Delta \partial_i u$ so that partials of u are also harmonic. Therefore,

$$\left|\frac{\partial u}{\partial x_i}(\mathbf{x}_0)\right| = \left|\frac{1}{B_{r/2}^{\mathbf{x}_0}}\int_{B_{r/2}^{\mathbf{x}_0}} \frac{\partial u}{\partial x_i}\,\mathrm{d}V\right| = \left|\frac{2^n}{\omega_n r^n}\oint_{\partial B_{r/2}^{\mathbf{x}_0}} u \cdot n_i \,\mathrm{d}S\right| \leq \frac{2n}{r}\sup_{\partial B_{r/2}^{\mathbf{x}_0}} |u|$$

For any $\mathbf{x} \in \partial B_{r/2}^{\mathbf{x}_0}$ we have $B_{r/2}^{\mathbf{x}} \subseteq B_r^{\mathbf{x}_0} \subseteq U$ and so from the $k=0$ case,

$$|u(\mathbf{x})| \leq \frac{1}{\omega_n}\left(\frac{2}{r}\right)^n \|u\|_{\mathrm{L}^1(B_r^{\mathbf{x}_0})}$$

Thus,

$$\frac{2n}{r}\sup_{\partial B_{r/2}^{\mathbf{x}_0}} |u| \leq \frac{2^{n+1}n}{\omega_n}\frac{1}{r^{n+1}}\|u\|_{\mathrm{L}^1(B_r^{\mathbf{x}_0})}$$

For all partials $\partial u/\partial x_i$. Now assume true up to $k-1$ and let $\boldsymbol{\alpha}$ be an k-tuple. By the same commutivity of partials with the Laplacian do we have $\partial_{\boldsymbol{\alpha}} u = \partial_i \partial_{\boldsymbol{\beta}} u$, $\boldsymbol{\beta}$ a $(k-1)$-tuple and $\partial_{\boldsymbol{\alpha}} u$ to be harmonic. Therefore by the mean-value property,

$$\left|\frac{\partial^{\boldsymbol{\alpha}} u}{\partial x_{\boldsymbol{\alpha}}}(\mathbf{x}_0)\right| = \left|\frac{1}{B_{r/k}^{\mathbf{x}_0}}\int_{B_{r/k}^{\mathbf{x}_0}} \frac{\partial^{\boldsymbol{\alpha}} u}{\partial x_{\boldsymbol{\alpha}}}\,\mathrm{d}V\right| = \left|\frac{k^n}{\omega_n r^n}\oint_{\partial B_{r/k}^{\mathbf{x}_0}} \frac{\partial^{\boldsymbol{\beta}} u}{\partial x_{\boldsymbol{\beta}}} \cdot n_i \,\mathrm{d}S\right| \leq \frac{kn}{r}\sup_{\partial B_{r/k}^{\mathbf{x}_0}} \left|\frac{\partial^{\boldsymbol{\beta}} u}{\partial x_{\boldsymbol{\beta}}}\right|$$

In a similar argument as the $k=1$ case if $\mathbf{x} \in \partial B_{r/k}^{\mathbf{x}_0}$ then $B_{(k-1)r/k}^{\mathbf{x}} \subseteq B_r^{\mathbf{x}_0} \subseteq U$ at which we may apply the inductive hypothesis to find,

$$\left|\frac{\partial^{\boldsymbol{\beta}} u}{\partial x_{\boldsymbol{\beta}}}(\mathbf{x})\right| \leq \frac{(2^{n+1}n(k-1))^{k-1}}{\omega_n((k-1)r/k)^{n+k-1}}\|u\|_{\mathrm{L}^1(B_r^{\mathbf{x}_0})}$$

And so in conjunction,

$$\frac{kn}{r}\sup_{\partial B_{r/k}^{\mathbf{x}_0}} \left|\frac{\partial^{\boldsymbol{\beta}} u}{\partial x_{\boldsymbol{\beta}}}\right| \leq \frac{(2^{n+1}nk)^k}{\omega_n r^{n+k}}\|u\|_{\mathrm{L}^1(B_r^{\mathbf{x}_0})}$$

■

Using the estimate we have the analogue of Liouville's Theorem in complex analysis.

THM 9. LIOUVILLE'S THEOREM

Let $u\colon \mathbb{R}^n \to \mathbb{R}$ be harmonic and bounded. Then u is constant.

Proof.

Let $\mathbf{x}_0 \in \mathbb{R}^n$ be arbitrary and we apply the Cauchy estimate of THEOREM 8.,

$$|\nabla u| \leq \frac{C_1}{r^{n+1}}\|u\|_{\mathrm{L}^1(B_r^{\mathbf{x}_0})} = \frac{C_1}{r^{n+1}}\int_{B_r^{\mathbf{x}_0}} |u| \,\mathrm{d}V \leq \frac{C_1|B_r^{\mathbf{x}_0}|}{r^{n+1}}\sup_{B_r^{\mathbf{x}_0}}|u| \leq \frac{C_1\omega_n}{r}\|u\|_{\mathrm{L}^\infty(\mathbb{R}^n)} \to 0$$

Hence $\nabla u = 0$, so u is constant.

■

Liouville's Theorem gives us a proof of the uniqueness of Poisson's equation on $\mathbb{R}^n$.

THM 10. REPRESENTATION THEOREM

Let $f \in \mathrm{C}^2_\mathsf{c}(\mathbb{R}^n)$ for $n \geq 3$. Then any bounded solution of the problem,

$$\begin{cases} -\Delta u = f, \quad \mathbf{x} \in \mathbb{R}^n \end{cases}$$

Is of the form,

$$u(\mathbf{x}) = \int_{\mathbb{R}^n} \Phi(\mathbf{x}-\mathbf{y})f(\mathbf{y})\,\mathrm{d}V_\mathbf{y} + C$$

For some constant C

Proof.

Note $\Phi(\mathbf{x}) \to 0$ as $|\mathbf{x}| \to 0$ and given f is compactly supported its convolution with Φ, i.e. u, is bounded. Moreover u solves the problem by THEOREM 2. Therefore let v be another solution to the problem and define $w := u - v$. Then w solves the problem,

$$\begin{cases} -\Delta w = 0, \quad \mathbf{x} \in \mathbb{R}^n \end{cases}$$

That is, is harmonic. Moreover the difference of two bounded functions is bounded, so w is bounded. By THEOREM 9. w must be constant $w = C$, so $v = u + C$.

■

This result does not hold for $n = 2$ as in that case Φ scales as $\ln|\mathbf{x}|$, which does decay to zero at infinity.

We now show that harmonic functions are more than infinitely-differentiable, but are also analytic, i.e. locally expressible as a power series.

THM 11. HARMONIC FUNCTIONS ARE ANALYTIC

Let u be harmonic on U. Then u is analytic on U.

Proof.

Let $\mathbf{x}_0 \in U$ be arbitrary and we find its power series about $\mathbf{x}_0$. Define $r := d(\mathbf{x}_0, \partial U)/4$ and $M := \|u\|_{\mathrm{L}^1(B_r^{\mathbf{x}_0})}/(\omega_n r^n)$. Given $B_r^{\mathbf{x}} \subseteq B_{2r}^{\mathbf{x}_0} \subseteq U$ for any $\mathbf{x} \in B_r^{\mathbf{x}_0}$ we get the Cauchy estimate,

$$\left|\frac{\partial^{\boldsymbol{\alpha}} u}{\partial x_{\boldsymbol{\alpha}}}(\mathbf{x})\right| \leq \frac{(2^{n+1}n|\boldsymbol{\alpha}|)^{|\boldsymbol{\alpha}|}}{\omega_n r^{n+|\boldsymbol{\alpha}|}}\|u\|_{\mathrm{L}^1(B_r^{\mathbf{x}})}$$

At which taking the supremum over all $\mathbf{x} \in B_r^{\mathbf{x}_0}$ yields, alongside Stirling's formula

$$\left\|\frac{\partial^{\boldsymbol{\alpha}} u}{\partial x_{\boldsymbol{\alpha}}}\right\|_{\mathrm{L}^\infty(B_r^{\mathbf{x}_0})} \leq M\left(\frac{2^{n+1}n}{r}\right)^{|\boldsymbol{\alpha}|}|\boldsymbol{\alpha}|^{|\boldsymbol{\alpha}|} \leq M\left(\frac{2^{n+1}n}{r}\right)^{|\boldsymbol{\alpha}|}Ce^{|\boldsymbol{\alpha}|}|\boldsymbol{\alpha}|! \leq CM\left(\frac{2^{n+1}n^2e}{r}\right)^{|\boldsymbol{\alpha}|}\prod_i(\alpha_i!)$$

The general Taylor series of multivariate function u about $\mathbf{x}_0$ is of the form,

$$u(\mathbf{x}_0) \approx \sum_{\boldsymbol{\alpha}} \frac{\partial^{\boldsymbol{\alpha}} u}{\partial x_{\boldsymbol{\alpha}}}|_{\mathbf{x}_0}\prod_i \frac{(x_i - x_{0,i})^{\alpha_i}}{\alpha_i!}$$

Therefore our task is to show that this sum converges. Consider the first Taylor expansion of the function $g(t) := u(\mathbf{x}_0 + t(\mathbf{x} - \mathbf{x}_0))$ about $t = 0$ up to N for,

$$\begin{aligned} g(t) &= \sum_{k=0}^{N-1}\frac{\partial^k g}{\partial t^k}|_0\frac{t^k}{k!} + \frac{1}{N!}\frac{\partial^N g}{\partial t^N}(t) \\ &= \sum_{|\mathbf{a}|\leq N-1}\frac{\partial^{\mathbf{a}} u}{\partial x_{\mathbf{a}}}|_{\mathbf{x}_0}\frac{t^k\prod_i(x_i - x_{0,i})^{\alpha_i}}{k!} + \frac{1}{N!}\sum_{|\mathbf{a}|=N}\frac{\partial^{\mathbf{a}} u}{\partial x_{\mathbf{a}}}(\mathbf{x}_0 + t(\mathbf{x} - \mathbf{x}_0))\prod_i(x_i - x_{0,i})^{\alpha_i} \end{aligned}$$

Namely by commutivity of partials the rearrangements of $\boldsymbol{\alpha}$ produce the same derivative. Thus we are concerned with the rearrangements of a k-array, at which we see the options are $C(k, \boldsymbol{\alpha})$. Hence to turn these partials into a multiindex $\mathbf{a} \to \boldsymbol{\alpha}$ we must group,

$$\sum_{|\mathbf{a}|=k}\frac{\partial^{\mathbf{a}} u}{\partial x_{\mathbf{a}}}|_{\mathbf{x}_0}\frac{t^k\prod_i(x_i - x_{0,i})^{\alpha_i}}{k!} = \sum_{|\boldsymbol{\alpha}|=k}\frac{\partial^{\boldsymbol{\alpha}} u}{\partial x_{\boldsymbol{\alpha}}}|_{\mathbf{x}_0}t^k\prod_i\frac{(x_i - x_{0,i})^{\alpha_i}}{\alpha_i!}$$

$$\frac{1}{N!}\sum_{|\mathbf{a}|=N}\frac{\partial^{\mathbf{a}} u}{\partial x_{\mathbf{a}}}(\mathbf{x}_0 + t(\mathbf{x} - \mathbf{x}_0))\prod_i(x_i - x_{0,i})^{\alpha_i} = \sum_{|\boldsymbol{\alpha}|=N}\frac{\partial^{\boldsymbol{\alpha}} u}{\partial x_{\boldsymbol{\alpha}}}(\mathbf{x}_0 + t(\mathbf{x} - \mathbf{x}_0))\prod_i\frac{(x_i - x_{0,i})^{\alpha_i}}{\alpha_i!}$$

At which at $t = 1$ we have,

$$u(\mathbf{x}) = g(1) = \sum_{|\boldsymbol{\alpha}|=k}\frac{\partial^{\boldsymbol{\alpha}} u}{\partial x_{\boldsymbol{\alpha}}}|_{\mathbf{x}_0}\prod_i\frac{(x_i - x_{0,i})^{\alpha_i}}{\alpha_i!} + \sum_{|\boldsymbol{\alpha}|=N}\frac{\partial^{\boldsymbol{\alpha}} u}{\partial x_{\boldsymbol{\alpha}}}(\mathbf{x})\prod_i\frac{(x_i - x_{0,i})^{\alpha_i}}{\alpha_i!}$$

The first term is precisely the $N - 1$ terms of the Taylor expansion of u about $\mathbf{x}_0$, so that with the equality with $u(\mathbf{x})$ we see that the latter term must be the remainder term R_N. This motivates choosing our radius to be,

$$|\mathbf{x} - \mathbf{x}_0| < \frac{r}{2^{n+2}n^3e}$$

At which,

$$\begin{aligned}
|R_N(\mathbf{x})| &= \left| \sum_{|\boldsymbol{\alpha}|=N} \frac{\partial^{\boldsymbol{\alpha}} u}{\partial x_{\boldsymbol{\alpha}}}(\mathbf{x}) \prod_i \frac{(x_i - x_{0,i})^{\alpha_i}}{\alpha_i!} \right| \\
&\leq \sum_{|\boldsymbol{\alpha}|=N} \left\| \frac{\partial^{\boldsymbol{\alpha}} u}{\partial x_{\boldsymbol{\alpha}}}(\mathbf{x}) \right\| \prod_i \frac{|x_i - x_{0,i}|^{\alpha_i}}{\alpha_i!} \\
&\leq CM \sum_{|\boldsymbol{\alpha}|=N} \left(\frac{2^{n+1} n^2 e}{r} \right)^{|\boldsymbol{\alpha}|} \left(\frac{r}{2^{n+2} n^3 e} \right)^N \\
&= CM \sum_{|\boldsymbol{\alpha}|=N} \frac{1}{(2n)^N} \\
&= \frac{CM}{2^N} && (|\{|\boldsymbol{\alpha}| = N\}| = n^N) \\
&\to 0
\end{aligned}$$

■